STEM LEARNING HUB

CODING FUNDAMENTALS — MODULE 1

Introduction to Programming

May 2, 2026

OVERVIEW

Learning Objectives

Define Programming

Understand programming as the process of creating instructions a computer can follow to perform tasks.

Importance of Coding

Recognize why coding is essential in today's digital world, powering apps, websites, and technology.

Write Your First Program

Learn to write and run a simple Python program, building a foundation for further coding skills.

Fundamentals

What Is Programming?

The Definition

Programming involves writing step-by-step instructions that a computer can understand and execute to perform specific tasks accurately and efficiently.

The Analogy

Think of programming like writing a recipe: you specify each step clearly, and the computer acts as the chef that follows your detailed instructions precisely.

Why It Matters

Without programming, we wouldn't have apps, websites, games, or the internet—code is the foundation powering all digital technology we use daily.

IMPORTANCE

Why Is Programming Important?

Mobile Apps

Every app on your smartphone, from social media to banking, relies on programming to function and deliver seamless user experiences.

Robotics

Robots operate through programmed instructions that allow them to perform complex tasks autonomously in manufacturing, healthcare, and more.

Artificial Intelligence

AI systems learn, adapt, and make decisions based on code, powering technologies like voice assistants, recommendation engines, and autonomous cars.

Video Games

Every game mechanic, character behavior, and interactive element is created through programming, making gaming immersive and dynamic.

Engineering Systems

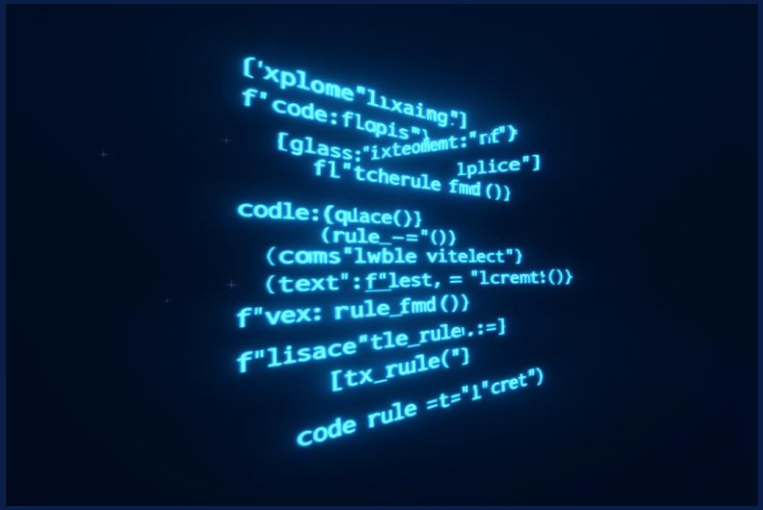
Critical infrastructure such as bridges, cars, and airplanes depend on embedded software to ensure safety, efficiency, and precision.

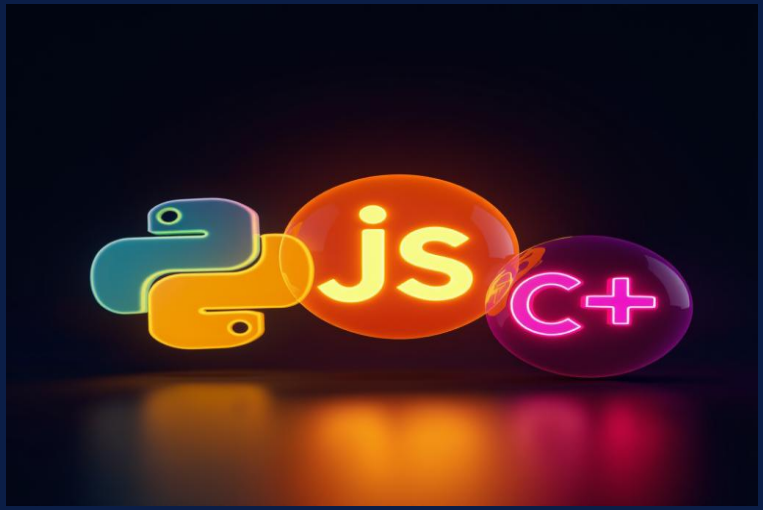
The Future

Programming is the foundational skill for tomorrow’s innovations, making it essential for future careers and technological advancement.

CONCEPTS

Programming Languages





What Is a Programming Language?

A programming language is a formal set of rules, syntax, and symbols that allows humans to write instructions a computer can understand and execute.

Popular Languages

Python is beginner-friendly and versatile; JavaScript powers web pages; C++ is used for systems software and games, each serving different programming needs.

How They Work

Humans write code, which is then translated by compilers or interpreters into machine language that the computer hardware can run.

PYTHON

Why Python?

Easy to Learn

Python's syntax is clean and reads almost like plain English, making it accessible and beginner-friendly for new programmers.

Incredibly Powerful

Python is used by top companies like Google, NASA, and Netflix for complex tasks, showcasing its strength and reliability.

Versatile & In-Demand

Python drives cutting-edge fields like AI, Data Science, Automation, and Web Development, offering vast career opportunities.

GETTING STARTED

Your First Python Program

```
1
1 ips Cetl {
2   print("Hello World")
3   print("Hello World")
4   }
5
```

Understanding the 'Hello, World!' Program

- The print() function tells the computer to display the text inside the quotes on the screen.
- Quotes " " define a string, which is a sequence of characters shown exactly as written.
- Running the program executes this line, producing the output 'Hello World' instantly in the console.

BASICS

Variables: Storing Information

Understanding Variables

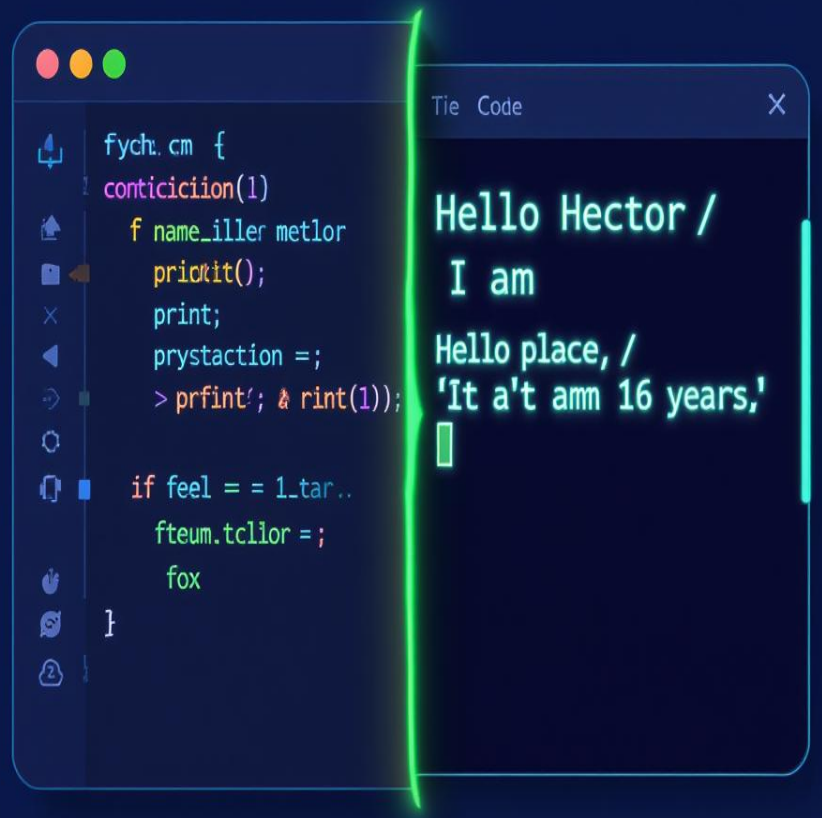
- A variable is like a labeled box that holds a value you can use and change later in your program.
- The variable 'name' stores text data, called a string, shown here as "Hector" in coral color.
- The variable 'age' stores a number, specifically an integer, shown here as 16 in teal color.
- Variables are flexible and their values can be updated anytime while the program runs.

OUTPUT

Displaying Output

How print() Combines Text and Variables

- print() can combine both text (strings) and variables in a single statement to show meaningful messages.
- Commas separate different items inside print(), letting Python insert spaces automatically between them.
- Variables like name and age are replaced with their current values when printed, creating dynamic output.
- The output appears instantly in the console, reflecting the exact content inside the print() function.



Inspiration

Real-Life Connection: What Can You Build?

A Game

Design characters, levels, and gameplay logic to bring your own video game ideas to life.

A Robot

Program physical machines to perform tasks, respond to sensors, and interact with environments.

A Website

Build interactive platforms and tools accessible worldwide that serve millions of users.

An App

Create mobile applications offering engaging experiences on smartphones and tablets.

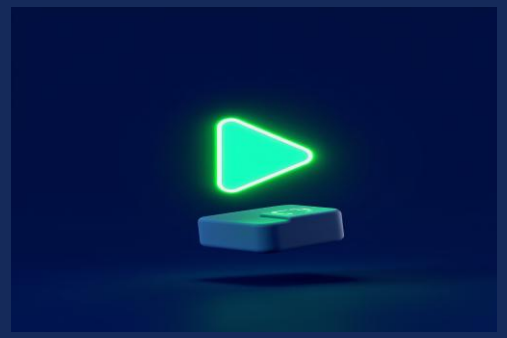
HANDS-ON PRACTICE

Practice Time: Google Colab

01

Open Google Colab

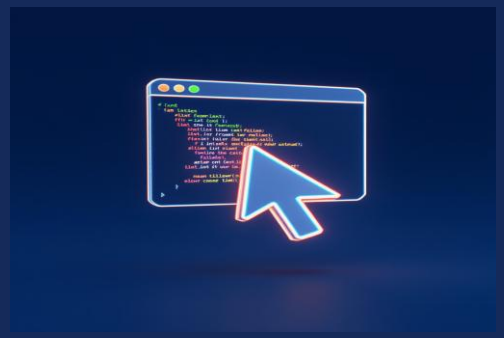
Navigate to colab.research.google.com and start a new notebook to begin coding in an interactive online environment.



02

Run the Starter Code

Type or paste the provided Python code snippet into a cell and click the Run button to execute the code and view output immediately.



03

Modify the Variables

Edit the name and age variables with your own information to personalize the program and see how changes affect output.

04

Add Your Favorite Technology

Create a new variable called favorite to store your favorite technology and use print() to display it on the screen.

PRACTICE

Practice Code

How to Use This Code

- Replace "Your Name" with your actual name to personalize the greeting.
- Update the age variable to reflect your current age as a number.
- Change the favorite variable to your favorite technology or interest.
- Run the entire code block in Google Colab to see your custom output appear instantly.

Common Mistakes to Avoid

Missing Quotes

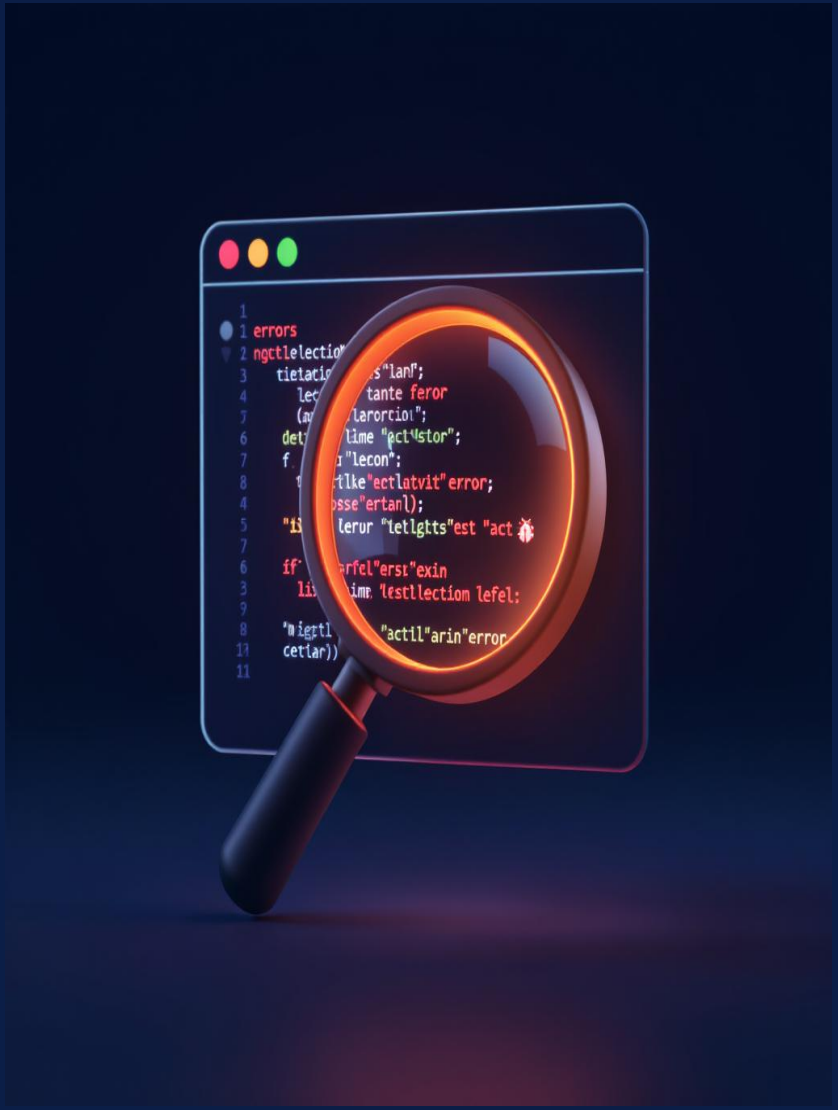
Strings must be enclosed in quotation marks " ". Forgetting quotes causes syntax errors that stop your code from running. Example: Wrong: print(Hello) vs. Right: print("Hello")

Misspelling print

Python is case-sensitive. Writing 'Print' or 'PRINT' instead of 'print' will cause errors because the function name must be lowercase.

Missing Parentheses

The print function requires parentheses. Omitting them leads to errors since Python won’t recognize the command without them. Example: print "Hello" is invalid; use print("Hello")



Reflection

01
What Is Programming?

Describe programming in your own words to someone new. Explain that it involves writing instructions a computer can follow to perform tasks.

02
What Did You Learn Today?

Identify one concept or skill from this module that surprised or excited you, such as writing your first Python program or understanding variables.

03
What Was Easy or Difficult?

Reflect on what parts of the lesson felt natural and which topics you found challenging. Consider what you want to focus on before moving to Module 2.

Up Next: Module 2

Module 2 will deepen your understanding of core data types and how to manipulate them to create dynamic programs.

Module 2 Preview

Module 2 focuses on Variables & Data Types, essential building blocks of programming. You'll explore integers, strings, floats, and booleans in detail, learning how to manipulate and transform data.

This knowledge lets you build more complex and dynamic programs that respond to user input and changing conditions.

Key Topics Covered

- Deep dive into integers, strings, floats, and booleans
- Learn how to manipulate and transform data effectively
- Build more complex and dynamic programs that think
- Apply new skills to create interactive, responsive code